Prop. Let V be any vector space, and $T: V \to V$ be a linear map. Suppose that there exists a T-invariant subspace W such that

$$V = \operatorname{Im} T \oplus W.$$

Then, $W \subseteq \ker T$.

Proof. Let $\alpha \in W$ be given. Then, $T(\alpha) \in \operatorname{Im} T \cap W$, so $T(\alpha) = 0$. Thus $T(\alpha) = 0$, so $\alpha \in \ker T$.

Prop. Let V be any vector space, and $T: V \to V$ be a linear map. Then

$$\operatorname{Im} T \cap \ker T = \{0\} \text{ if and only if } \ker T = \ker T^2.$$

Proof. Notice that in general, $\ker T \subseteq \ker T^2$.
Suppose that $\operatorname{Im} T \cap \ker T = \{0\}$. Let $\alpha \in \ker T^2$ be given. Then, $T^2(\alpha) = T(T(\alpha)) = 0$, so that $T(\alpha) \in \operatorname{Im} T \cap \ker T$, so $T(\alpha) = 0$. Thus $\alpha \in \ker T$. Conversely, suppose that $\ker T = \ker T^2$. Given $\alpha \in \operatorname{Im} T \cap \ker T$, $\alpha = T(\beta)$ for some $\beta \in V$ and $T(\alpha) = 0$. Thus, $T(\alpha) = T^2(\beta) = 0$, so $\beta \in \ker T^2 = \ker T$. Therefore $\alpha = T(\beta) = 0$. So, proved. ☐

Thm. Let V be a finite-dimensional vector space, let $T: V \to V$ be linear. There is $k \in \mathbb{N}$ such that $V = \operatorname{Im} T^k \oplus \ker T^k$.

Proof. Since $T(0) = 0$, so given $\alpha \in \ker T$, $T^2(\alpha) = T(0) = 0$, thus $\alpha \in \ker T^2$. Inductively, we obtain a chain

$$\ker T \subseteq \ker T^2 \subseteq \cdots \subseteq V$$

But, since the V is finite-dimensional, for some $k \in \mathbb{N}$ such that $k \leq \dim V$,

$$\ker T^{k-1} \subsetneq \ker T^k = \ker T^{k+1} = \cdots$$

Let $\alpha \in \operatorname{Im} T^k \cap \ker T^k$ be given. Then, $\alpha = T^k(\beta)$ for some $\beta \in V$, and so that $0 = T^k(\alpha) = T^{2k}(\beta)$, so $\beta \in \ker T^{2k} = \ker T^k$, thus $\alpha = T^k(\beta) = 0$. Now, by the dimension theorem, there exist a basis $\mathcal{B}_1$ for $\operatorname{Im} T^k$ and $\mathcal{B}_2$ for $\ker T^k$ such that

$$|\mathcal{B}_1| + |\mathcal{B}_2| = \dim V,$$

And $\mathcal{B}_1 \cup \mathcal{B}_2$ is linearly independent, since $\langle \mathcal{B}_1 \rangle \cap \langle \mathcal{B}_2 \rangle = \operatorname{Im} T^k \cap \ker T^k = \{0\}$, so, Consequently $V = \operatorname{Im} T^k \oplus \ker T^k$. ∎